Phase 6d report: LFP transfer on the BattGP field data

Data: Zenodo 13715694, `field_data.zip` (1,668,409,464 bytes, MD5 verified against Zenodo's 827f5087...), CC BY-NC 4.0 (research only), downloaded 2026-11-03 to `data/battgp_lfp/` (made read-only; never modified). Contents: 28 CSV files (`data_sys_1..28.csv`), about 19.9 GB uncompressed, 133 million rows per the Zenodo README, 24 V / about 160 Ah LFP systems, 8 cells in series, one current sensor, 9 voltage sensors, 4 temperature sensors, active balancing. All systems were returned to the manufacturer for unsatisfactory behaviour, so the sample is biased (not representative of the fleet). The BattGP cache (feather) and results live in `data/battgp_cache/` and `data/battgp_results/` (git-ignored).

6d.1 correctness check against the paper: PASSED on cell identification and timing, with caveats

Paper read: arXiv 2406.19015v3 (the preprint; it says it is outdated, the published version may differ). `python -m sunfix.lfp_reproduce 3 9 6` runs BattGP unchanged (recursive spatiotemporal GP, CPU, 1-hour step). Three settings were run: `repo` (BattGP's gp_runner defaults: fixed reference operating point -15 A / 90% SOC / 25 degC, band 0.55 mOhm, current limit -80 A, SOC up to 95%), `paper` (paper Table 3 criteria: current -200..-5 A, SOC 40-94%, band b = 0.33 mOhm), `paper_meanref` (paper settings with the mean-of-data reference operating point, which the paper describes). One system at a time: system 3 13 s, system 9 36 s (3.7 GB peak), system 6 2.3 min (6.5 GB peak). What the paper says vs what we get: | Paper (arXiv v3) | This reproduction | |---|---| | System 3: "cell 7 shows a strong upward resistance trend, while other cells have fairly constant resistances" (127-day section) | Cell 7 has by far the largest resistance rise (+1.4 mOhm against 0.2-0.4 for the others, paper settings); same under repo settings (+0.46 vs about 0.1) | | System 6: "cell 8 ... fault probability increases after 600 days and crosses 0.5 shortly before 1000 days"; small probability at the start | paper_meanref, smoothed: cell 8 about 0.01 at the start, first above 0.5 on day 1020; cell 8 rises +1.1 mOhm vs at most 0.12 for the others | | System 9: "cell 2 ... knee point after 240 days" (Table 2; its text says day 220 and, confusingly, system 10) | Cell 2 first above 0.5 on day 241 (paper_meanref) / 234-241 (other settings); rise +2.6 to 3.2 mOhm vs at most 0.3 | Caveats, all material: - **The reference operating point matters.** With BattGP's default fixed reference point (repo setting), system 6 cell 8 is flagged for the whole period (probability 1.0 from the start), which contradicts the paper's "small at the start"; with the mean-of-data reference it matches. Any pack-only comparison must fix this choice up front. - **Absolute-threshold probability (2 mOhm) is not in the paper text** I could read, and saturates at 1.0 for every cell and system here because each cell's R0 at the reference point is about 8-9 mOhm. Treat it as an unvalidated repo feature and ignore it. - **Forward (causal) probabilities have a start-up transient** (about 0.5-1.0 in the first days while the filter variance is large). Onset times must exclude a burn-in (30 days used below). - Paper's own figure caption and panel titles disagree about system numbers, and the text calls the system with the day-220 knee "system 10"; the Table 2 row for system 9 is used here. - Hyperparameters: BattGP's shipped LFP values (medians from the paper's runs); the paper tuned on systems 6, 8, 9, 10. - Paper numbers for causal probabilities, exact curves and the figures were not compared (figures not read).

Reconciliation

- **Systems, rows:** the paper (arXiv v3) says 29 systems, 232 cells and 131 million rows ("29 battery systems returned to the manufacturer for warranty"). The Zenodo release has **28 systems (28 CSV files)** and says 133 million rows, so the discrepancy is between the paper and the released data (the paper's 29 is probably the count before one system was dropped from the release; this is a guess). BattGP's `gp_runner.py` loops over IDs 1-29. The paper models 21 of its systems ("latest section with no gaps larger than 100 days and at least 600 valid data points"). The "28" in the paper is a count of basis vectors, not systems.

- **Sampling rate:** the paper reports a median measurement interval of 5 s, matching what we measured on systems 3 and 9 (median 5 s; 79-85% of intervals at most 5 s; the rest mostly 60 s; duplicated timestamps 0.1-0.2%). Filter updates use a 1-hour sampling time.

6d.2 Key experiment: does a pack-only signal detect the single-cell faults? Answer: not demonstrated (inconclusive, leaning negative)

Code: `sunfix/lfp_pack.py`, `lfp_pack_run.py`, `lfp_sections.py`. Pack voltage = sum of the 8 cell voltages; per-cell voltages, balancing currents and BMS SOC are dropped. OCV-free resistance: for each discharge run (current below -2 A; observations below -5 A) the reference is the median pack voltage of the rest samples (|I| < 1 A) in the preceding 5 minutes, R = (V_rest - V) / |I|; the resistance GP (Wiener-velocity time process + SE over current and temperature, per-run random offset, causal filter) is the same machinery as the lead-acid work. Same analysis windows as BattGP's per-cell run (`results/lfp_section_times.csv`). **Pseudo-labels** (`results/lfp_percell_pseudolabels.csv`). The paper-literal weakest-link probability (band 0.33 mOhm, mean reference) flags 19 of 21 systems from day 30 because early-life cell spread already exceeds the band, so it cannot say when a fault starts. We therefore define a fault as a cell whose resistance relative to the other cells rises by at least 0.33 mOhm above its own days 30-60 baseline, sustained 7 days (smoothed trajectories). 21 systems are modelled (22-28 fail BattGP's minimum-data filter, matching the paper's 21). Eligible 19 (section at least 90 days); system 1 excluded (implausible 125 mOhm drift, data artefact); **10 fault systems, 8 controls** (max drift 0.4 mOhm or less); systems 5 and 12 too short. **Result.** At a +20% pack-resistance alarm (causal, sustained 7 days): 2 of 10 fault systems detected (alarm no earlier than 30 days before the per-cell onset), 2 further alarms more than 30 days early, 2 of 8 controls falsely flagged. This changes a lot with the label-free hyperparameter fit set: refitting on three different random sets of four systems gives 2, 4 and 0 of 10 detected, 2, 6 and 0 of 8 controls falsely flagged, and a threshold-free AUC (end-of-section over baseline resistance, faults vs controls) of 0.54, 0.17 and 0.92 (`results/lfp_pack_only_robustness.csv`, figure `lfp_pack_only.png`). **Conclusion: with 10 faults and 8 controls and this method, pack-level data cannot be shown to detect the cell faults; the evidence is inconclusive, leaning negative.** Why we expect it to be hard: the fitted per-run offset of the rest-voltage reference is 18-55 mOhm, larger than the pack resistance itself (about 9-50 mOhm), and a single cell contributes only a part of the pack resistance. **Honest process note.** The fit bounds, the alarm rule (CUSUM, then a relative-rise rule) and the labels were revised several times while looking at these same 19 systems; no result here is a clean held-out test. The first CUSUM version false-flagged 6 of 8 controls (the GP baseline noise was too small); a bound pinning the offset hyperparameter (a lead-acid unit setting) was found and fixed.

6d.3 Degraded pack-only telemetry (`results/minimum_data_table_lfp.csv`)

Same degradations as Phase 4 applied to the pack-only model (fixed seed-0 hyperparameters; no-temperature refits). Because the baseline detection is unstable, the **detection columns are not reliable**; the coverage columns are. Systems (of 18 labelled) with enough pack observations for a baseline: native 17, 1-minute 17, 5-minute 10, 15-minute 7, 1-hour 8 (median observations per system 1,960, 996, 250, 139, 21); no temperature 16; history limits, voltage rounding to 0.1 V, current rounding to 1 A and packet loss up to 50% 17 each. Detection stays at 0-2 of 10 and false flags at 0-7 of 8 in every scenario, i.e. noise around a method that does not separate the groups. The firm statement is observability: the rest-voltage method loses about half of its systems at 5-minute sampling and cannot be baselined reliably at 15 minutes or longer (the rest-lookback and minimum-run settings were scaled with the interval to be fair).

6d.4 Lead-acid to LFP transfer (`results/lfp_transfer_rank_agreement.csv`)

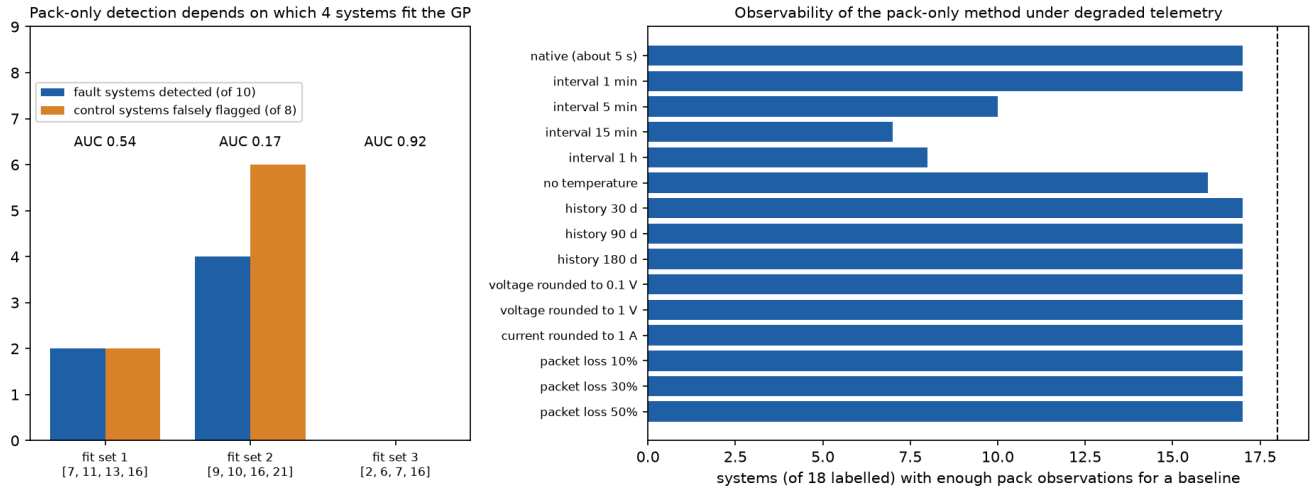
A LightGBM trained on Bboxx landmarks with only baseline-relative resistance features (level over baseline; slopes over 7/30/90 days over baseline; age and usage dropped because they do not transfer between 12 V lead-acid and 24 V LFP) reaches only AUC 0.66 on its own at-risk test cohort. Applied to the pack-only LFP

trajectories (11 of the 19 eligible systems have enough data), its predicted risk (mean over the last 30 days) has **Spearman correlation -0.34 with the per-cell maximum drift (95% bootstrap interval -0.74 to +0.33, p = 0.31) and AUC 0.50 for faults against controls: no rank agreement.**

What 6d establishes

1. BattGP reproduces on the Zenodo data (cell identification and onset timing match the paper under its settings; the reference operating point matters).
2. We could not show that cheap pack-level telemetry detects single-cell LFP faults; the evidence is inconclusive, leaning negative, with fragile, small-sample results.
3. A lead-acid model does not rank LFP systems like the per-cell method.
4. For LFP the safe conclusion for an open data specification is to keep per-cell voltage (or at least per-cell min/max summaries) in the spec until a better pack-level method is demonstrated on a larger labelled data set.

Figures



lfp_pack_only.png